Big Data Analysis of E-commerce Reviews and Price Trends in High-End Digital and Gaming Products

Group H

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ABSTRACT

The goal of this project is to appraise e - commerce reviews and price fluctuations of high - end digital and gaming products with big data technologies.

The study will amass a significant amount of data from digital platforms like Amazon and Kaggle, including reviews of products, ratings, and historical prices. PySpark will be used to handle the data, which will then be stored in distributed databases such as HDFS and MongoDB.

In addition to that, sentiment analysis and trend analysis will be conducted by means of Spark MLlib. Well - structured charts are used to present the results. The project shows that big data can be used to support data - driven decision - making.

1.TOPIC SELECTION

When shopping in the high - end digital and esports hardware market, consumers frequently face complex buying decisions. Flagship smartphones, GIGABYTE liquid - cooling radiators, and professional esports peripherals are expensive and technically sophisticated, causing difficulty for consumers to make a choice.

A large amount of unstructured data is generated as a result of e - commerce platforms. Though this data is useful for trend evaluation, it's hard to process using traditional equipment.

The project uses big data tech to look into user sentiment and price trends. By integrating review analysis and pricing data, it intends to find the connection between consumer feedback and market trends.

The results of this study will help consumers make more judicious buying decisions and help companies improve pricing strategies and product development.

2. CORE ISSUES TO BE ADDRESSED

This project will focus on the fields of high-end digital and e-sports peripherals, and the analysis will focus on three core issues:

（1） Price configuration correlation analysis: How do different configuration parameters (such as storage capacity, memory size, processor model) affect product pricing strategy and price volatility? Are there phenomena such as "configuration premium" and "cost-effective inflection point"?

（2） User emotion and feedback mining: What product dimensions do consumers value most when buying flagship smartphones or e-sports devices? Is there a significant difference in user satisfaction between different configuration versions?

（3） Differences in brand and platform performance: What is the pattern of change in the price trend of the same product at different times (such as before and after the release of new products and during the promotion season)? Is there a systematic deviation in the distribution of user evaluations of different brands in the same product category?

3. RELEVANCE TO COURSE AND TECHNICAL SOLUTIONS

3.1 Phase 1: Data Collection

This project plans to use open datasets to build a data foundation:

（1） Data source: The project plans to utilize Kaggle's public dataset Amazon Electronics Reviews, which contains over 500,000 authentic user reviews. The dataset includes fields such as product name, rating, review text, and timestamp, making it suitable for large-scale text analysis and sentiment mining.

（2） Standard Data Fields:Product ID, Product Name, Configuration Version, Price, Review Content, Rating, Review Time, Brand, Platform Source.

3.2 Phase 2: Data preprocessing and cleaning

The specific data cleaning process is as follows:

a. Missing value processing method: eliminate the record that the comment content is empty or the score is missing; for samples with missing price fields, the average value of the same model and the configured product is used for interpolation calculation.
b. Duplicate record deduplication: Identifies and removes duplicate comments based on the combination of "user ID + product ID + comment time" to prevent bias in sentiment analysis results caused by duplicate data.
c. Data type conversion: standardize the note time field into yyyy-mm-dd format; convert the price field into numerical type; and extract product configuration information from the structure.
d. Text cleaning: perform basic operations on the comment text, including disabling specific words, removing special symbols and unifying the case format, so as to provide high-quality input data for subsequent emotional analysis and word cloud construction.
e. Abnormal detection: Identify the price abnormal value through box diagram and distribution analysis, and then carry out manual evaluation to determine the retention standard.

3.3 Phase3: Storage and Processing of Data

We will utilize the hybrid distributed system to handle the vast amount of e-commerce data; its heterogeneity:

HDFS: A system that will be deployed can reliably store massive, structured raw pricing and transactional data in different distributed chunkservers.

Mongo DB (NoSQL): It will be used in order to save semi-structured and unstructured consumer reviews. Its document model based data structure will provide a flexible schema and multi-dimensional storage, and can store complex data association without the inflexibility of classical RDBMS.

PySpark and Spark SQL: It will implement Spark SQL and PySpark as the central processing unit because of its rapid, in-memory cluster computing applications. Using the DataFrame API (RDD + schema) backed by a Parquet Data Lake (Parquet column-format files) to store intermediate data, that will be able to use as many distributions of SQL queries and multi-dimensional aggregations that could be efficiently scaled, such as average price drop across various memory configurations.

3.4 Phase4: Data Visualization and analytics

In order to formulate raw data into precise findings and identify the market trends:

Spark MLlib: It will be used to perform sentiment analysis of the review texts, which will be stored in the MongoDB. Through its machine learning classification models, which are scalable like the Logistic Regression or Linear SVM, we will classify the user feedback in order to reveal the particular pain points of the high-end digital products.

Data Visualization: Finished data will be transferred to Python tools as Matplotlib or Seaborn. Our intuitively created visual summaries such as price trend line charts, sentiment distributions among others will enable easy consumer understanding of complex trends.

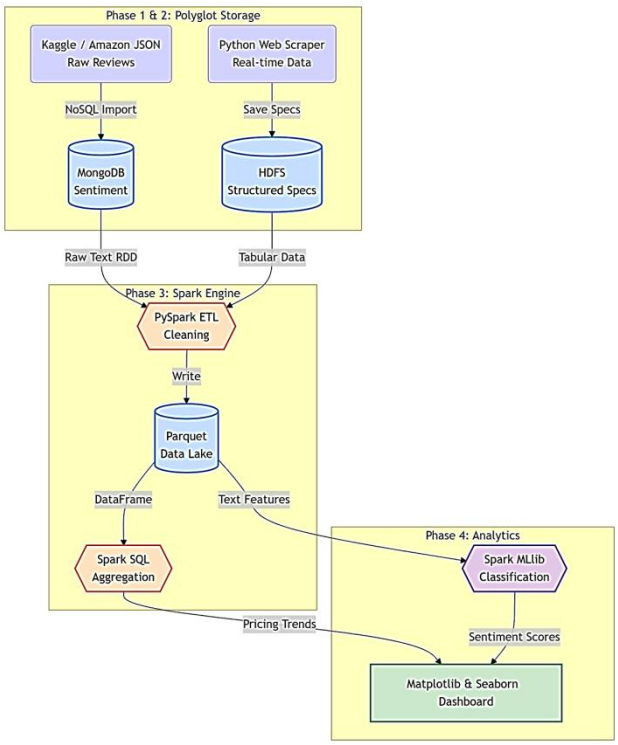


Figure 3.1 Four Phases of Our Project (tentative)

4. GROUP DIVISION OF LABOR

To make sure of the efficient execution of this big - data project and the technical architecture in Figure 3.1, the workload has been logically divided among five team members, which directly ties to the data flow and analytical nodes of our technical architecture. Each member will be responsible for specific technical implementations and analytical goals

Team Member 1: Data Collection & Architecture Specialist

Responsibilities: In charge of building a sound data foundation. We make use of Kaggle's public dataset for historical Amazon Electronics Reviews and develop a Python Web Scraper to bring in real - time e - commerce data. Will roll out and control the polyglot distributed storage system, taking care of HDFS for structured product details (e.g. , leading smartphone and esports peripheral configurations) and MongoDB for unstructured review texts.

Team Member 2: Data Quality Engineer Responsibilities: Focuses on the data preprocessing, quality assurance, and the ETL pipeline execution. The tasks include handling missing values, removing duplicate entries based on "user ID + product ID + comment time", and conducting text cleaning (eliminating special symbols and filtering stop - words). Significantly, this role is to carry out the process of anomaly detection. Box - and - whisker plots will be used to manually appraise and filter away price outliers. Finally, this member is in charge of the PySpark ETL process to put the refined data into the Parquet Data Lake for optimized downstream queries.

Team Member 3: Big Data Analyst

Responsibilities: In charge of processing the Spark engine. Who will make use of PySpark and the DataFrame API to run distributed SQL queries and do multi - dimensional aggregations. This role should attempt to tackle the first core issue: analyzing the correlation between price and configuration and assessing phenomena such as "configuration premium".

Team Member 4: Machine Learning Engineer Responsibilities: Controls the Phase 4 analytics through the use of Spark MLlib. Who is to train scalable machine - learning classification models, say Logistic Regression or Linear SVM, for sentiment analysis of user reviews. This member concentrates on the second core issue.

Team Member 5: Visualization Expert & Project Coordinator

Responsibilities: The job is to transfer the completed analytical data to Python tools such as Matplotlib or Seaborn. Who is going to put up clear visual summaries, like price trend charts and sentiment distributions. The member acting as the coordinator will blend the findings to address the third core issue related to brand and platform performance differences.

5. NEXT STEPS AND PROJECT PLAN

Step 1: Build the environment and bring data in. We are going to finish setting up our distributed cluster environments to ensure proper integration of HDFS, MongoDB, and Spark. At the same time, we'll extract over 500,000 real user reviews and structural data fields (e.g. , Product ID, Price, Rating) gotten from our data sources.

Step 2:Execution of the ETL Pipeline. Guided by the Data Quality Engineer, the team will run the initial PySpark data - cleaning scripts to make time formats standard (yyyy-mm-dd) and change price fields to numeric types. We will carry out anomaly detection in a systematic way through box and whisker plots to check price outliers. Only high - quality tabular and text data will be taken into the Parquet Data Lake for downstream analytics.

Step 3: Advanced Analytics and Model Training. We are going to do Spark SQL multi - dimensional aggregations to measure the average price drops across different configurations. At the same moment, we will introduce the pre - processed review data into Spark MLlib to develop strong sentiment scores and reveal particular pain points of high - end digital products.

Step 4: Establish Dashboards and Extract Findings. The accumulated pricing trends and sentiment scores will be imported into Matplotlib and Seaborn to build an all - in - one dashboard. We will visualize the biases in user reviews in a systematic way and find pricing inflection points.

Step 5: Execute Final Evaluation and Craft Report. In the final part, we're going to map the visualized data back to the three main problems. We will put together our findings to give practical advice to businesses about optimizing pricing and product - development strategies, proving big data is useful for decision - making.